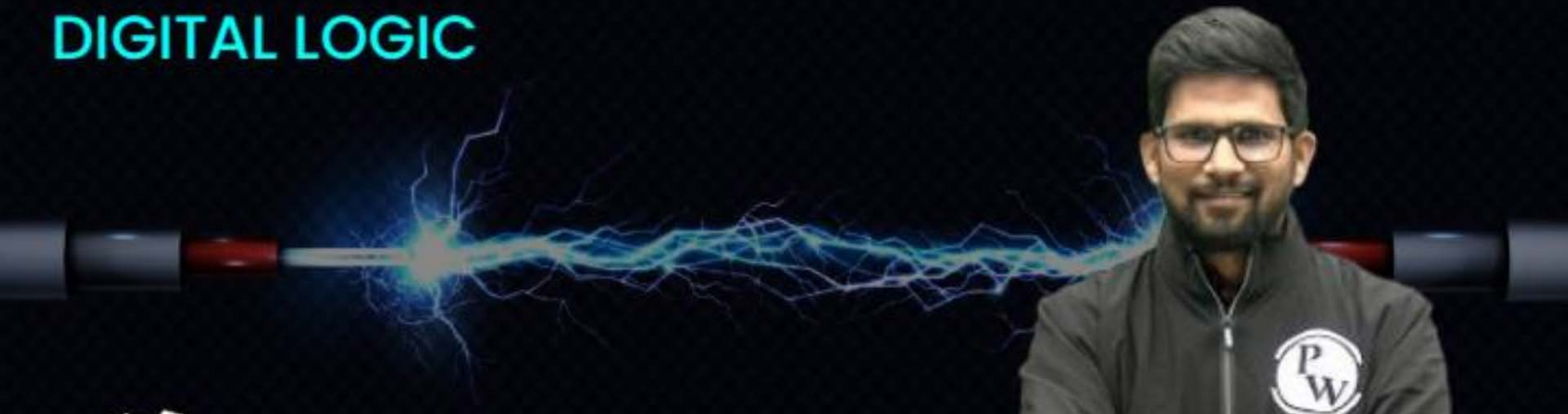
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 04

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture

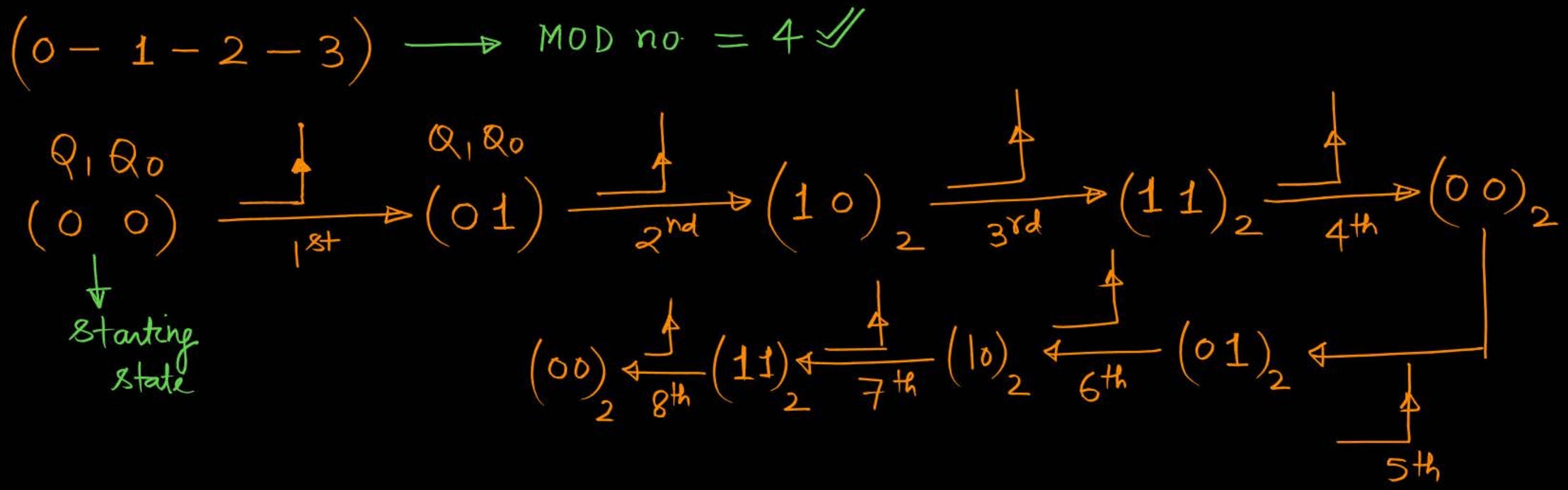


- Race - Around condition
- Counter
-

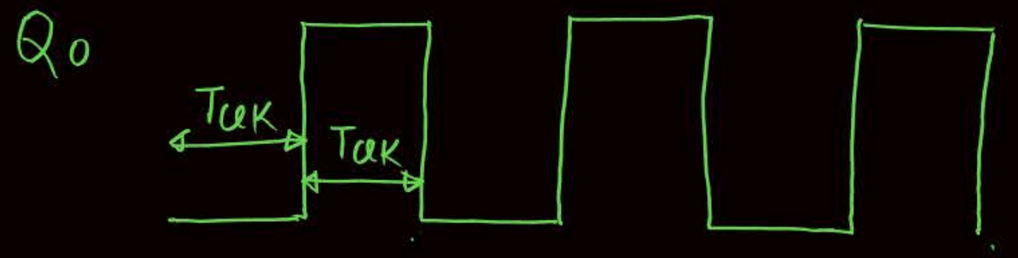


Counter Cont.

- MOD No. of the counter :



Q_1	Q_0
0	0
0	1
1	0
1	1



$Q_0 \rightarrow T_{off} = T_{CLK}$

$T_{ON} = T_{CLK} \longrightarrow T_{Q_0} = 2T_{CLK}$

$f_{Q_0} = \frac{f_{CLK}}{2}$

$Q_1 \rightarrow T_{off} = 2T_{CLK}$

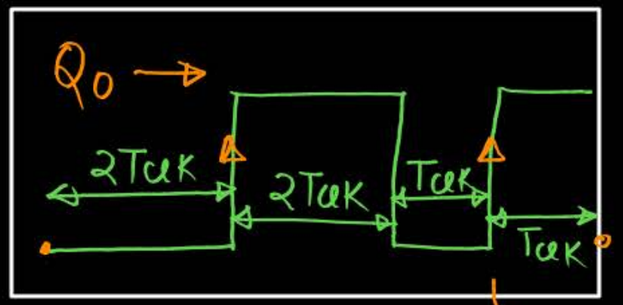
$T_{ON} = 2T_{CLK}$

$T_{Q_1} = 4T_{CLK} = T_{ON} + T_{off}$

$f_{Q_1} = f_{CLK}/4$

- $(0-2-3-1-4-5)$ MOD no. $= 6$

	Q_2	Q_1	Q_0
	0	0	0
1st ↑	0	1	0
2nd ↑	0	1	1
3rd ↑	0	0	1
4th ↑	1	0	0
5th ↑	1	0	1



$T_{Q_0} = 6T_{CLK}$

$f_{Q_0} = f_{CLK}/6$

$Q_1 \rightarrow$ $T_{OFF} = 4T_{CLK}$, $T_{ON} = 2T_{CLK}$ $\rightarrow$ $T_{Q_1} = 6T_{CLK}$

$f_{Q_1} = \frac{f_{CLK}}{6}$

$Q_2 \rightarrow$ $T_{OFF} = 4T_{CLK}$, $T_{ON} = 2T_{CLK}$ $\rightarrow$ $T_{Q_2} = 6T_{CLK}$

$f_{Q_2} = f_{CLK}/6$

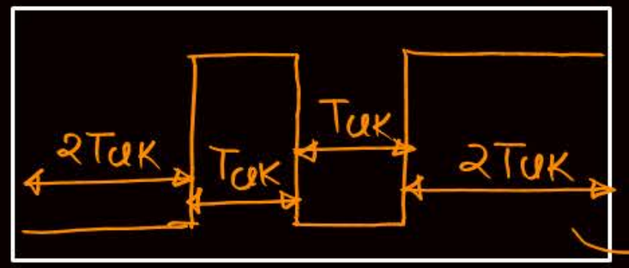
- $(0 - 3 - 4 - 7 - 6 - 5)$

MOD no. $= 6$

Q_2	Q_1	Q_0
0	0	0
0	1	1
1	0	0
1	1	1
1	1	0
1	0	1

$Q_0 \rightarrow T_{off} = 1T_{clk}$, $T_{ON} = 1T_{clk} \longrightarrow T_{Q_0} = 2T_{clk}$, $f_{Q_0} = f_{clk}/2$

$Q_1 \rightarrow$



→ disjoint waveform

$T_{Q_1} = 6T_{clk}$

$f_{Q_1} = f_{clk}/6$

$Q_2 \rightarrow T_{off} = 2T_{clk}$, $T_{ON} = 4T_{clk} \longrightarrow T_{Q_2} = 6T_{clk}$, $f_{Q_2} = f_{clk}/6$

→ starting state $- (4)_{10}$

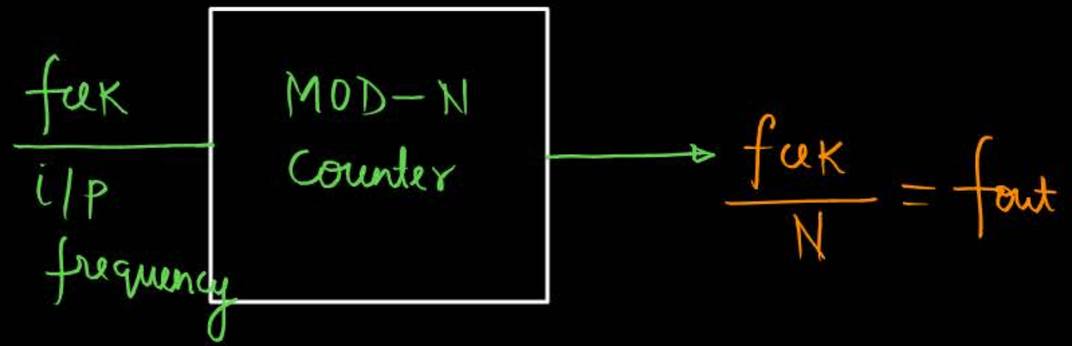
After 197 clock pulses the state of the counter will be $(3)_{10}$

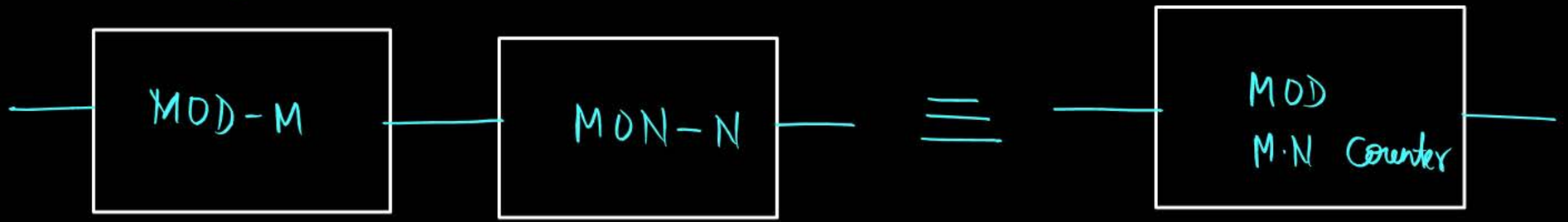
$(4)_{10} \xrightarrow{6\,clk} (4)_{10} \xrightarrow{6\,clk} (4)_{10} \xrightarrow{6\,clk} (4)_{10}$ ⋮ $(4)_{10} \xrightarrow[192\,clk]{32 \times 6} (4)_{10} \xrightarrow{5\,clk} (3)_{10}$

- Properties related to MOD No. :

- After MOD no. of clock pulses counter goes to its starting state.

- After application of integer multiple of MOD no. of clock pulses, counter will be again at its starting state.

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Types of Counter

Asynchronous Counter

- Different ffs are driven by different clocks.
- Only fixed sequence is possible
  - up sequence: 0-1-2-3-4...
  - down sequence: 7-6-5-4-3-2
- Relatively more delay and therefore asynchronous counters are relatively slow.

→ Ex. BCD counter, any fixed sequence

Synchronous Counter

- All the ffs are driven by same clock signal.
- Any sequence is possible.
- Relatively less delay and therefore relatively faster.

Eg: Ring Counter, Johnson Counter etc.
Any Random sequence, Any fixed sequence.

- No. of ffs required for designing a counter

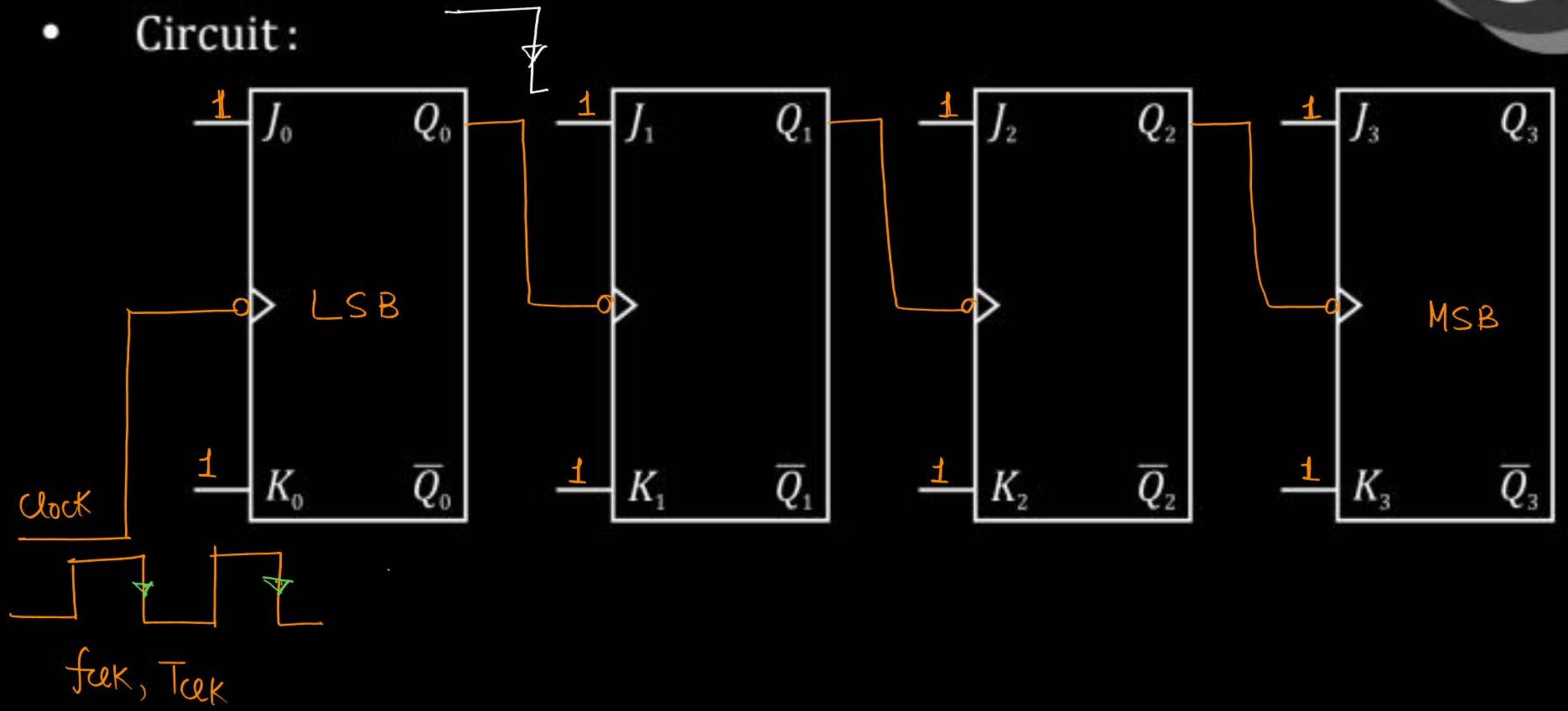
MOD no. = M

n - ffs for designing this counter.

$2^n \geq M$ ✓✓

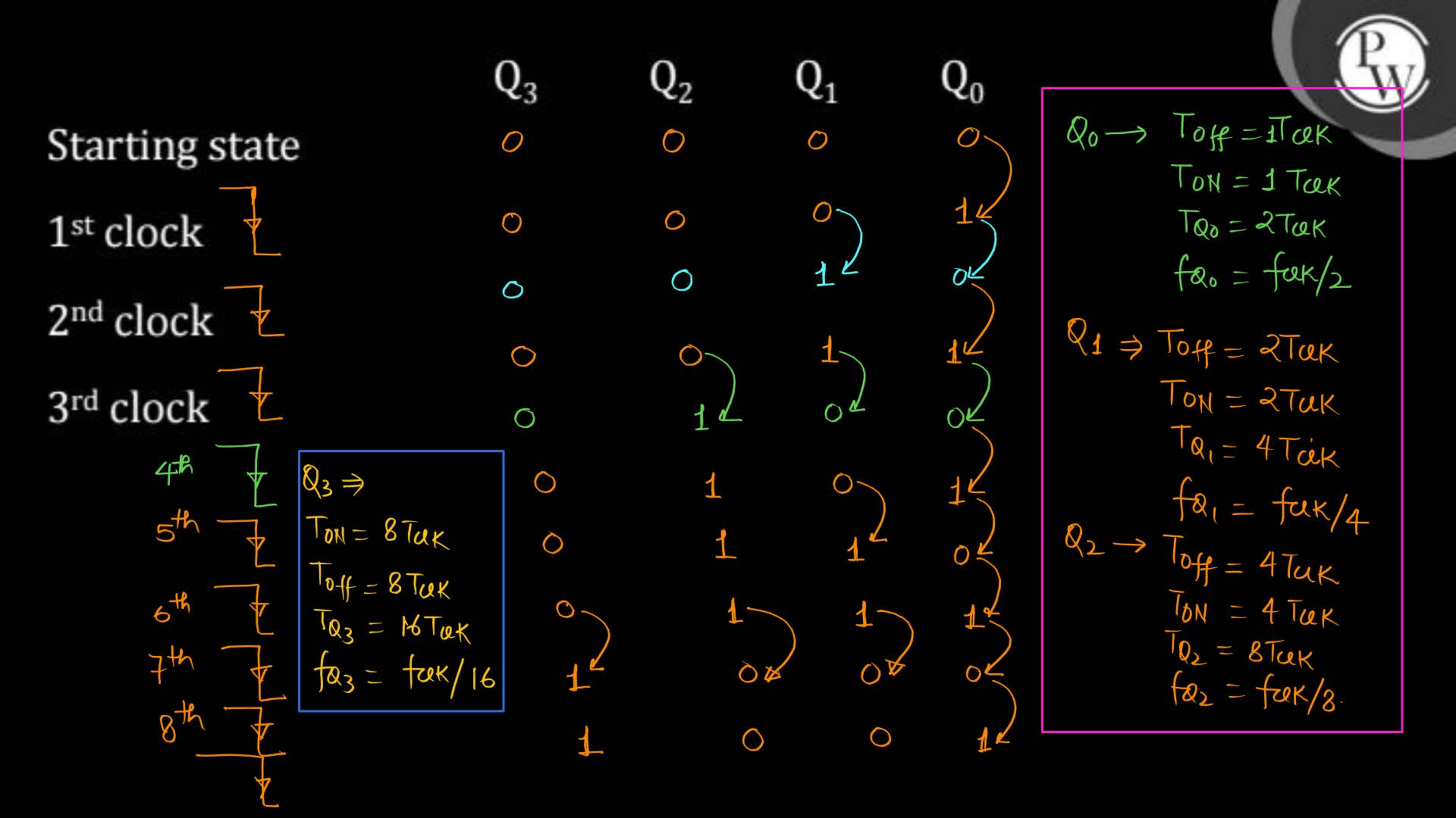
4-bit Asynchronous Counter

- Circuit :



- Working :

- Q_0 toggles in every clock.
- Q_1 toggles when Q_0 take transition from (1-0).
- Q_2 toggles when Q_1 take transition from (1-0)
- Q_3 toggles when Q_2 take transition from (1-0)


Starting state
1st clock
2nd clock
3rd clock
4th
5th
6th
7th
8th
Q3 Q2 Q1 Q0
Q3 ⇒
TON = 8 Tclk
Toff = 8 Tclk
TQ3 = 16 Tclk
fQ3 = fclk/16
Q0 → Toff = 1 Tclk
TON = 1 Tclk
TQ0 = 2 Tclk
fQ0 = fclk/2
Q1 ⇒ Toff = 2 Tclk
TON = 2 Tclk
TQ1 = 4 Tclk
fQ1 = fclk/4
Q2 → Toff = 4 Tclk
TON = 4 Tclk
TQ2 = 8 Tclk
fQ2 = fclk/8.

10th 1 0 1 0

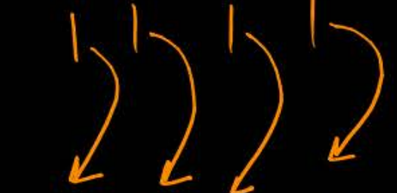
11th 1 0 1 1

12th 1 1 0 0

13th 1 1 0 1

14th 1 1 1 0

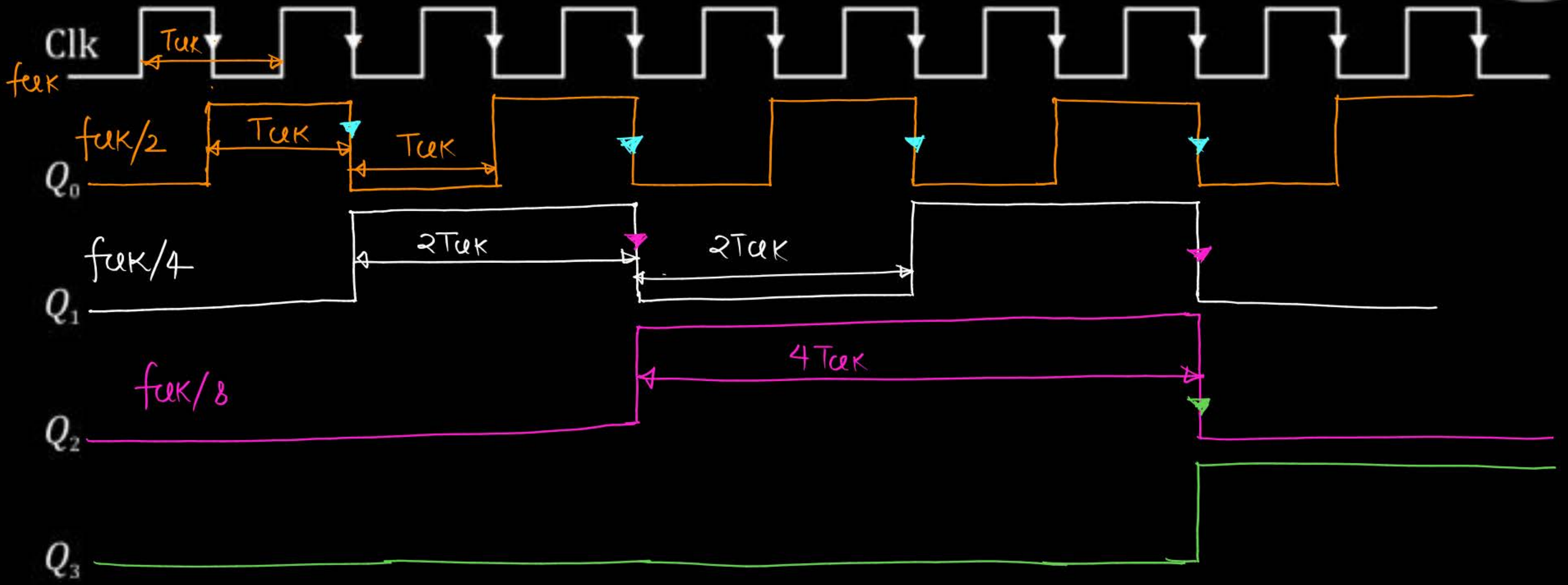
15th 1 1 1 1



16th clock (0 0 0 0)

MOD No. = 16

- Clock Diagram :





Topic : 2 Min Summary

Asynch. Counter

Thank you

GW

Soldiers !

